

Q1 JEE Main 2020 - 7 Jan (Morning)

Let $A(1, 0)$, $B(6, 2)$ and $C\left(\frac{3}{2}, 6\right)$ be the vertices of a triangle ABC . If P is a point inside the triangle ABC such that the triangles APC , APB and BPC have equal areas, then the length of the line segment PQ , where Q is the point $\left(-\frac{7}{6}, -\frac{1}{3}\right)$, is.....

Q2 JEE Main 2020 - 7 Jan (Evening)

The locus of the mid-point of the perpendiculars drawn from points on the line, $x = 2y$ to the line $x = y$ is:

- (A) $2x = 3y$
- (B) $5x = 7y$
- (C) $3x = 2y$
- (D) $7x = 5y$

Q3 JEE Main 2020 - 8 Jan (Morning)

Let two points be $A(1, -1)$ and $B(0, 2)$. If a point $P(x', y')$ be such that the area of $\triangle PAB = 5$ square units and it lies on the line, $3x + y - 4\lambda = 0$, then a value of λ is :

- (A) 3
- (B) -3
- (C) 4
- (D) 2

Q4 JEE Main 2020 - 8 Jan (Evening)

The length of the perpendicular from the origin on the normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at the point $(2, 2)$ is

- (A) $4\sqrt{2}$
- (B) 2
- (C) $2\sqrt{2}$
- (D) 4

Q5 JEE Main 2020 - 9 Jan (Morning)

$A(3, -1)$, $B(1, 3)$, $C(2, 4)$ are vertices of $\triangle ABC$ if D is centroid of $\triangle ABC$ and P is point of intersection of lines $x + 3y - 1 = 0$ and $3x - y + 1 = 0$ then which of the following points lies on line joining D and P

- (A) $(-9, -7)$
(B) $(-9, -6)$
(C) $(9, 6)$
(D) $(9, -6)$

Q6 JEE Main 2020 - 2 Sep (Evening)

The set of all possible values of θ in the interval $(0, \pi)$ for which the points $(1, 2)$ and $(\sin \theta, \cos \theta)$ lie on the same side of the line $x + y = 1$ is :

- (A) $(0, \frac{\pi}{2})$
(B) $(\frac{\pi}{4}, \frac{3\pi}{4})$
(C) $(0, \frac{\pi}{4})$
(D) $(0, \frac{3\pi}{4})$

Q7 JEE Main 2020 - 3 Sep (Evening)

If a $\triangle ABC$ has vertices $A(-1, 7)$, $B(-7, 1)$ and $C(5, -5)$, then its orthocentre has coordinates

- (A) $(-3, 3)$
(B) $(-\frac{3}{5}, \frac{3}{5})$
(C) $(3, -3)$
(D) $(\frac{3}{5}, -\frac{3}{5})$

Q8 JEE Main 2020 - 4 Sep (Morning)

A triangle ABC lying in the first quadrant has two vertices as $A(1, 2)$ and $B(3, 1)$. If $\angle BAC = 90^\circ$, and $\text{ar}(\triangle ABC) = 5\sqrt{5}$ sq. units, then the abscissa of the vertex C is :

- (A) $1 + \sqrt{5}$
(B) $1 + 2\sqrt{5}$
(C) $2\sqrt{5} - 1$
(D) $2 + \sqrt{5}$

Q9 JEE Main 2020 - 4 Sep (Evening)

If the perpendicular bisector of the line segment joining the points $P(1, 4)$ and $Q(k, 3)$ has y -intercept equal to -4 , then a value of k is:

(A) $\sqrt{14}$

(B) $\sqrt{15}$

(C) -4

(D) -2

Q10 JEE Main 2020 - 5 Sep (Morning)

If the line, $2x - y + 3 = 0$ is at a distance $\frac{1}{\sqrt{5}}$ and $\frac{2}{\sqrt{5}}$ from the lines $4x - 2y + \alpha = 0$ and $6x - 3y + \beta = 0$, respectively, then the sum of all possible values of α and β is

Q11 JEE Main 2020 - 6 Sep (Morning)

A ray of light coming from the point $(2, 2\sqrt{3})$ is incident at an angle 30° on the line $x = 1$ at the point A. The ray gets reflected on the line $x = 1$ and meets x-axis at the point B. Then, the line AB passes through the point:

(A) $\left(3, -\frac{1}{\sqrt{3}}\right)$

(B) $(3, -\sqrt{3})$

(C) $(4, -\sqrt{3})$

(D) $\left(4, -\frac{\sqrt{3}}{2}\right)$

Q12 JEE Main 2020 - 6 Sep (Evening)

Let L denote the line in the xy-plane with x and y intercepts as 3 and 1 respectively. Then the image of the point $(-1, -4)$ in this line is :

(A) $\left(\frac{29}{5}, \frac{8}{5}\right)$

(B) $\left(\frac{29}{5}, \frac{11}{5}\right)$

(C) $\left(\frac{8}{5}, \frac{29}{5}\right)$

(D) $\left(\frac{11}{5}, \frac{28}{5}\right)$

Answer Key**Q1** (5)**Q2** (B)**Q3** (A)**Q4** (C)**Q5** (B)**Q6** (A)**Q7** (A)**Q8** (B)**Q9** (C)**Q10** (30)**Q11** (B)**Q12** (D)

Q1

P will be centroid of ΔABC

$$P\left(\frac{17}{6}, \frac{8}{3}\right) \Rightarrow PQ = \sqrt{\left(\frac{24}{6}\right)^2 + \left(\frac{9}{3}\right)^2} = 5$$

Q2

$$\text{slope of } PQ = \frac{k - \alpha}{h - 2\alpha} = -1$$

$$\Rightarrow k - \alpha = -h + 2\alpha$$

$$\Rightarrow \alpha = \frac{h + k}{3} \quad \dots (1)$$

$$\text{Also } 2h = 2\alpha + \beta$$

$$2k = \alpha + \beta$$

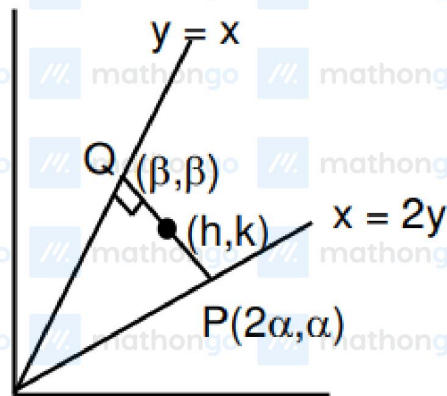
$$\Rightarrow 2h = \alpha + 2k$$

$$\Rightarrow \alpha = 2h - 2k \quad \dots (2)$$

from (1) and (2)

$$\frac{h + k}{3} = 2(h - k)$$

$$\text{so locus } 6x - 6y = x + y \Rightarrow 5x = 7y$$



Q3

$$D = \frac{1}{2} \begin{vmatrix} 0 & 2 & 1 \\ 1 & -1 & 1 \\ x' & y' & 1 \end{vmatrix}$$

$$-2(1 - x') + (y' + x') = \pm 10$$

$$-2 + 2x' + y' + x' = \pm 10$$

$$3x' + y' = 12 \text{ or } 3x' + y' = -8$$

$$\lambda = 3, -2$$

Q4

$$x^2 + 2xy - 3y^2 = 0$$

$$x^2 + 3xy - xy - 3y^2 = 0$$

$$(x - y)(x + 3y) = 0$$

$$x - y = 0 \quad x + 3y = 0$$

$$(2, 2) \text{ satisfy } x - y = 0$$

Normal

$$x + y = \lambda$$

$$\lambda = 4$$

$$\text{Hence } x + y = 0$$

$$\text{perpendicular distance from origin} = \left| \frac{0 + 0 - 4}{\sqrt{2}} \right| = 2\sqrt{2}$$

Q5

$$D(2, 2)$$

$$\text{Point of intersection } P\left(-\frac{1}{5}, \frac{2}{5}\right)$$

equation of line DP

$$8x - 11y + 6 = 0$$

Q6

$$\text{Let } f(x, y) = x + y - 1$$

$$\therefore f(1, 2) \cdot f(\sin \theta, \cos \theta) > 0$$

$$\Rightarrow 2[\sin \theta + \cos \theta - 1] > 0$$

$$\Rightarrow \sin \theta + \cos \theta > 1$$

$$\Rightarrow \sin\left(\theta + \frac{\pi}{4}\right) > \frac{1}{\sqrt{2}}$$

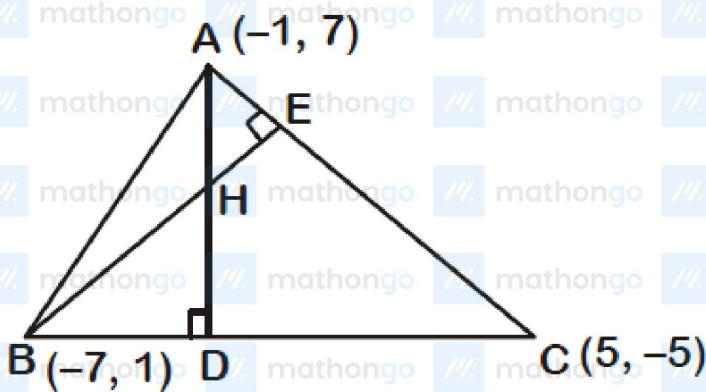
$$\Rightarrow \theta + \frac{\pi}{4} \in \left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$$

$$\Rightarrow \theta \in \left(0, \frac{\pi}{2}\right)$$

Q7

$$\text{Here } m_{BC} = \frac{-6}{12} = \frac{-1}{2}$$

$$\text{So } m_{AD} = 2$$



So equation of $AD \Rightarrow y - 7 = 2(x + 1)$

$$2x - y + 9 = 0$$

Now

$$m_{AC} = \frac{12}{-6} = -2 \Rightarrow m_{BE} = \frac{1}{2}$$

So equation of $BE \Rightarrow (y - 1) = \frac{1}{2}(x + 7)$

$$\Rightarrow 2y - 2 = x + 7 \Rightarrow x - 2y + 9 = 0$$

On solving (i) and (ii)

$$2x - y + 9 = 0$$

$$x - 2y + 9 = 0$$

$$2x - y + 9 = 0$$

$$2x - 4y + 18 = 0$$

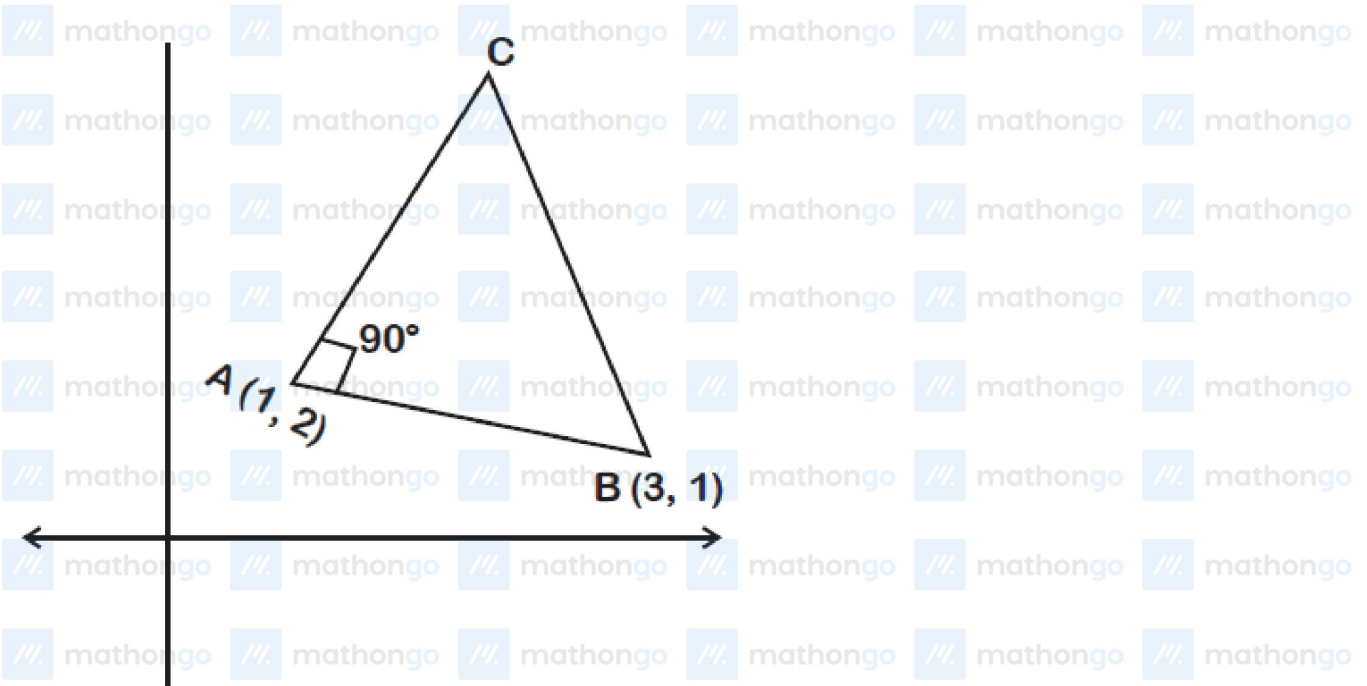
$$3y - 9 = 0$$

$$\Rightarrow y = 3$$

$$\Rightarrow x = -3$$

So $H = (-3, 3)$

Q8



$$\text{Slope of line } AB = -\frac{1}{2}$$

$$\text{Slope of line } AC = 2$$

$$\text{Length of } AB = \sqrt{5}$$

$$\therefore \frac{1}{2} AB \cdot AC = 5\sqrt{5}$$

$$\therefore AC = 10$$

$$\therefore \text{Coordinate of } C = (1 + 10 \cos \theta, 2 + 10 \sin \theta)$$

$$\text{Here } \tan \theta = 2 \Rightarrow \cos \theta = \frac{1}{\sqrt{5}}, \sin \theta = \frac{2}{\sqrt{5}}$$

$$\text{Coordinate of } C = (1 + 2\sqrt{5}, 2 + 4\sqrt{5})$$

$$\text{abscissa of vertex } C \text{ is } 1 + 2\sqrt{5}$$

Q9

$$\text{Mid point of line segment PQ is } \left(\frac{k+1}{2}, \frac{7}{2} \right)$$

$$\text{Slope of PQ is } \frac{1}{1-k}$$

So equation of perpendicular bisector of PQ is

$$y - \frac{7}{2} = (k-1) \left[x - \frac{k+1}{2} \right]$$

$\therefore$ Line passes through (0,-4); then

$$-\frac{15}{2} = -\frac{(k^2-1)}{2} \Rightarrow k = \pm 4$$

Q10

$$L_1 : 2x - y + 3 = 0$$

$$L_1 : 4x - 2y + \alpha = 0$$

$$L_1 : 6x - 3y + \beta = 0$$

Distance between L_1 and L_2

$$\left| \frac{\alpha - 6}{2\sqrt{5}} \right| = \frac{1}{\sqrt{5}} \Rightarrow |\alpha - 6| = 2$$

$$\Rightarrow \alpha = 4, 8$$

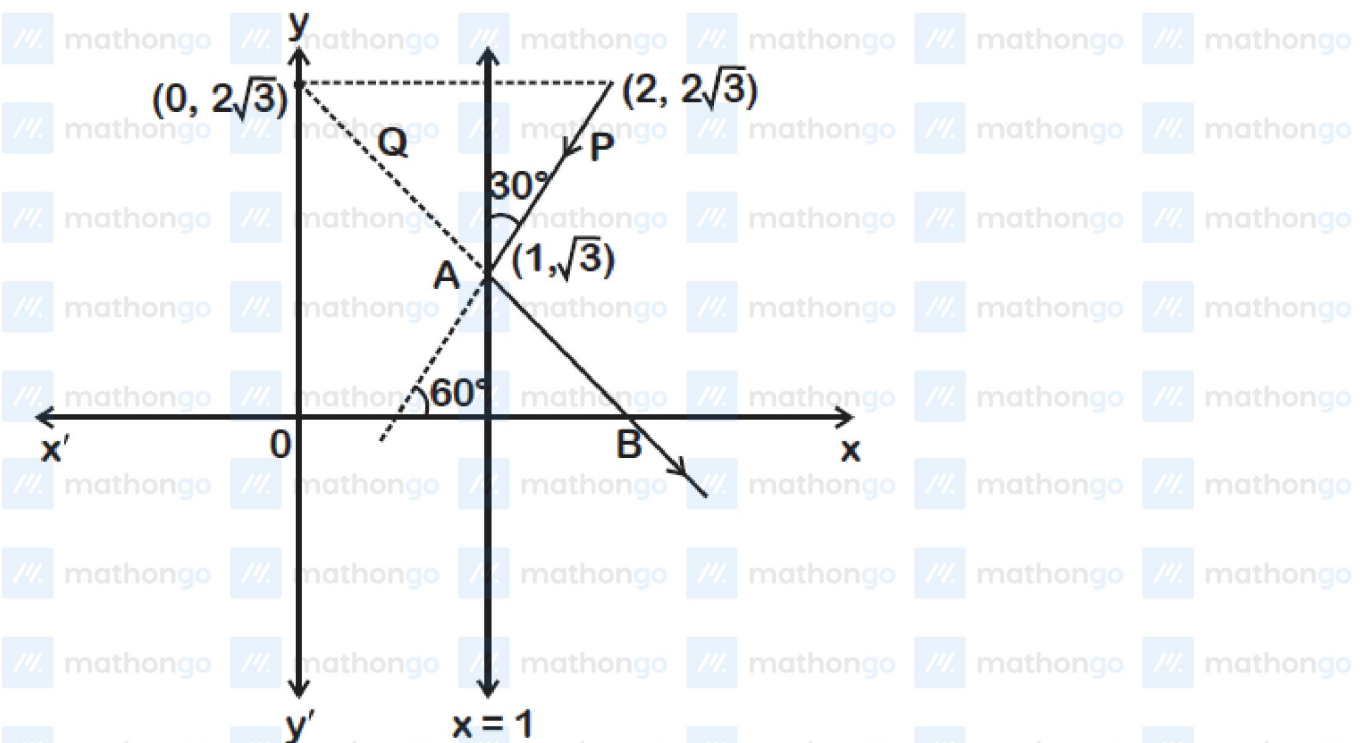
$$\text{Distance between } L_1 \text{ and } L_3 \left| \frac{\beta - 9}{3\sqrt{5}} \right| = \frac{2}{\sqrt{5}} \Rightarrow |\beta - 9| = 6$$

$$\Rightarrow \beta = 15, 3$$

$$\text{Sum of all values} = 4 + 8 + 15 + 3 = 30$$

Q11

Equation of incident line AP is



$$y - 2\sqrt{3} = \sqrt{3}(x - 2)$$

$$\sqrt{3}x - y = 0$$

Image of P w.r.t. line $x = 1$

is point $Q = (0, 2\sqrt{3})$

Equation of reflected Ray AB :

$$y - \sqrt{3} = \frac{2\sqrt{3} - \sqrt{3}}{0 - 1}(x - 1)$$

$$\sqrt{3}x + y = 2\sqrt{3}$$

$\therefore$ Point $(3, -\sqrt{3})$ lies on line AB

Q12

Line is $\frac{x}{3} + y = 1$

$$x + 3y - 3 = 0$$

Let image be (α, β)

$$\text{Hence, } \frac{\alpha+1}{1} = \frac{\beta+y}{3} = \frac{2(-1-12-3)}{10}$$

$$\alpha + 1 = \frac{\beta+4}{3} = \frac{16}{5}$$

$$\Rightarrow \alpha = \frac{11}{5}, \beta = \frac{28}{5}$$